

# DATA SCIENCE

CONTINUOUS ASSESSMENT - 4

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# Agenda

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## Topics Covered

1

INTRODUCTION

2

DATASET DESCRIPTION & EDA

3

SIMPLE LINEAR REGRESSION

4

MULTIPLE LINEAR REGRESSION

5

K-MEANS

2

DECISION TREE

7

LOGISTIC REGRESSION

8

CONCLUSION

# INTRODUCTION

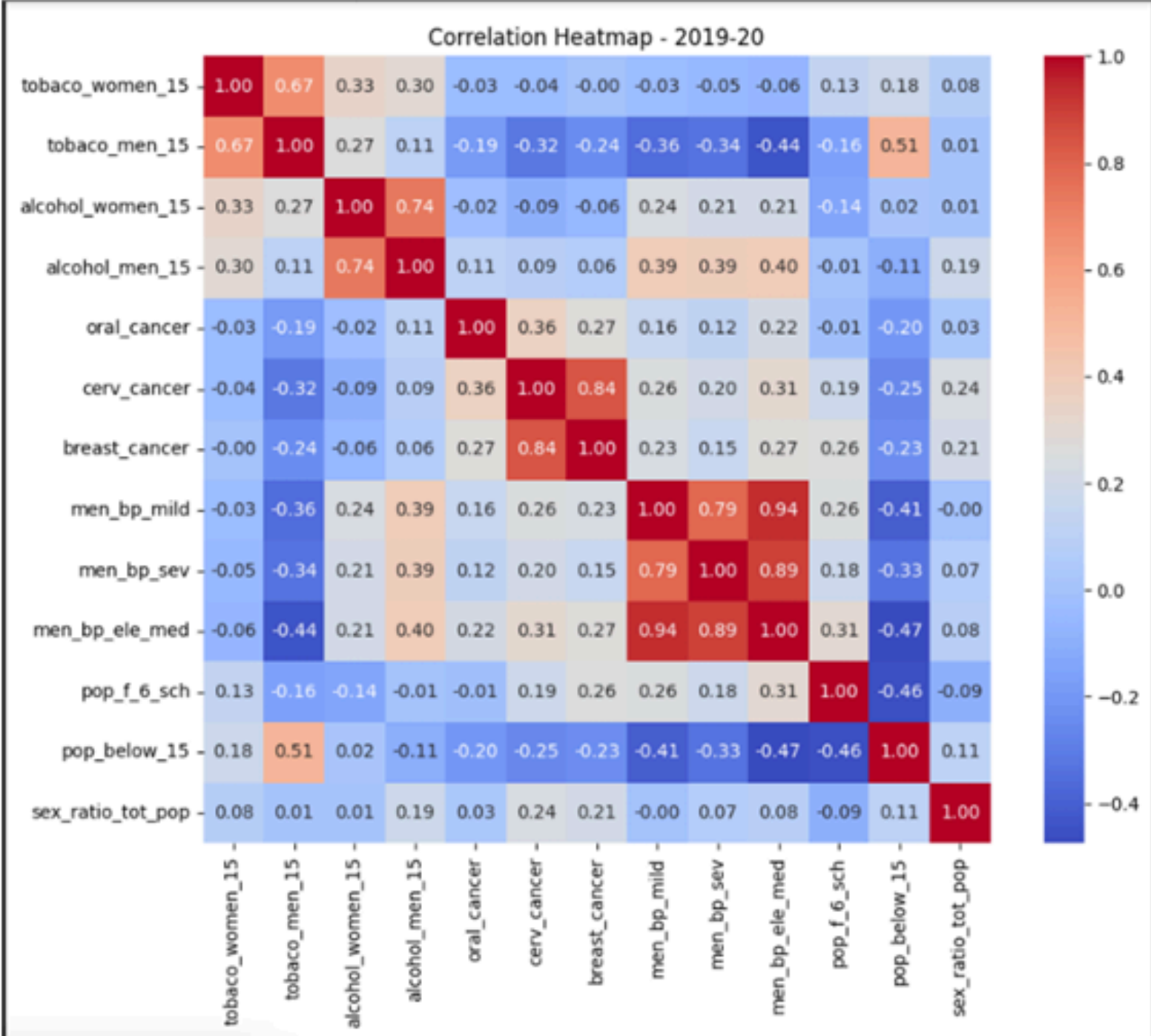
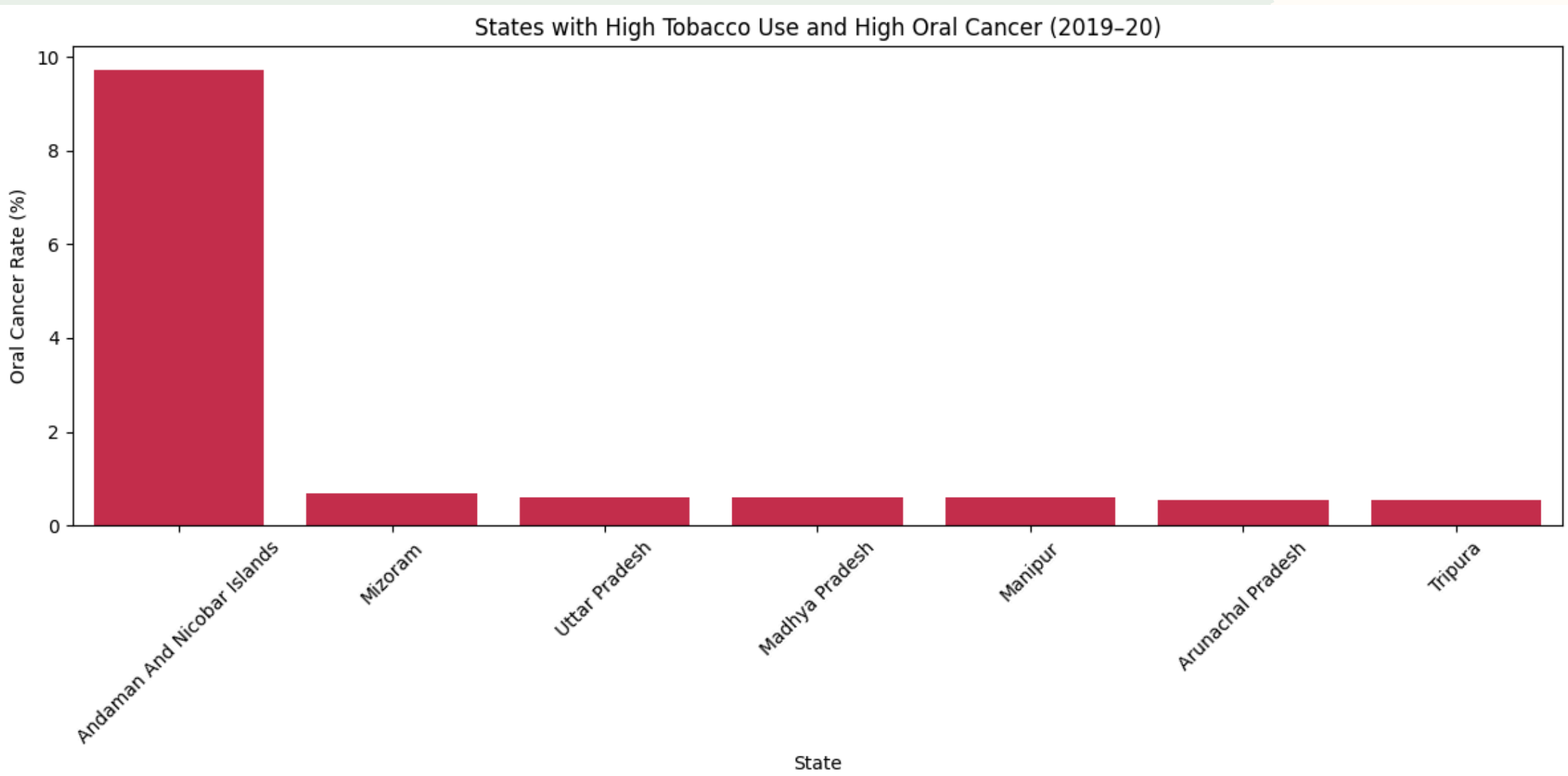
What the Project really is about

- This project analyzes data from the National Family Health Survey from India Data Portal to explore the socio-demographic, behavioral, and health factors influencing teenage pregnancy and women's health outcomes in India.
- Using statistical and machine learning models, including linear regression, clustering, and decision trees ,the study identifies patterns linking education, early marriage, anaemia, and nutrition to women's empowerment and reproductive health.
- The dataset covers multiple indicators such as population structure, family planning, child health, and chronic diseases across Indian states and districts.
- I chose this topic because it reflects a real and pressing public-health issue, allowing data science to uncover actionable insights.

# DATASET DESCRIPTION

- Source: India Data Portal
- Time Period: July 2014 – March 2015 and July 2019 – March 2020
- Scope: Nationwide – all Indian states and union territories
- Observations: 1,300 district-level records
- Features: 110 variables, with 15 key indicators selected for focused analysis
- Dataset Link : <https://indiadataportal.com/p/national-family-health-survey/r/mohfw-nfhs-dt-qg-eie>

# EDA FINDINGS





# Analytical Models and Methodology Overview

## SIMPLE LINEAR REGRESSION

A simple linear regression was applied to analyze how early marriage (% women married before 18) affects teenage pregnancy (% girls aged 15–19 pregnant) across districts.

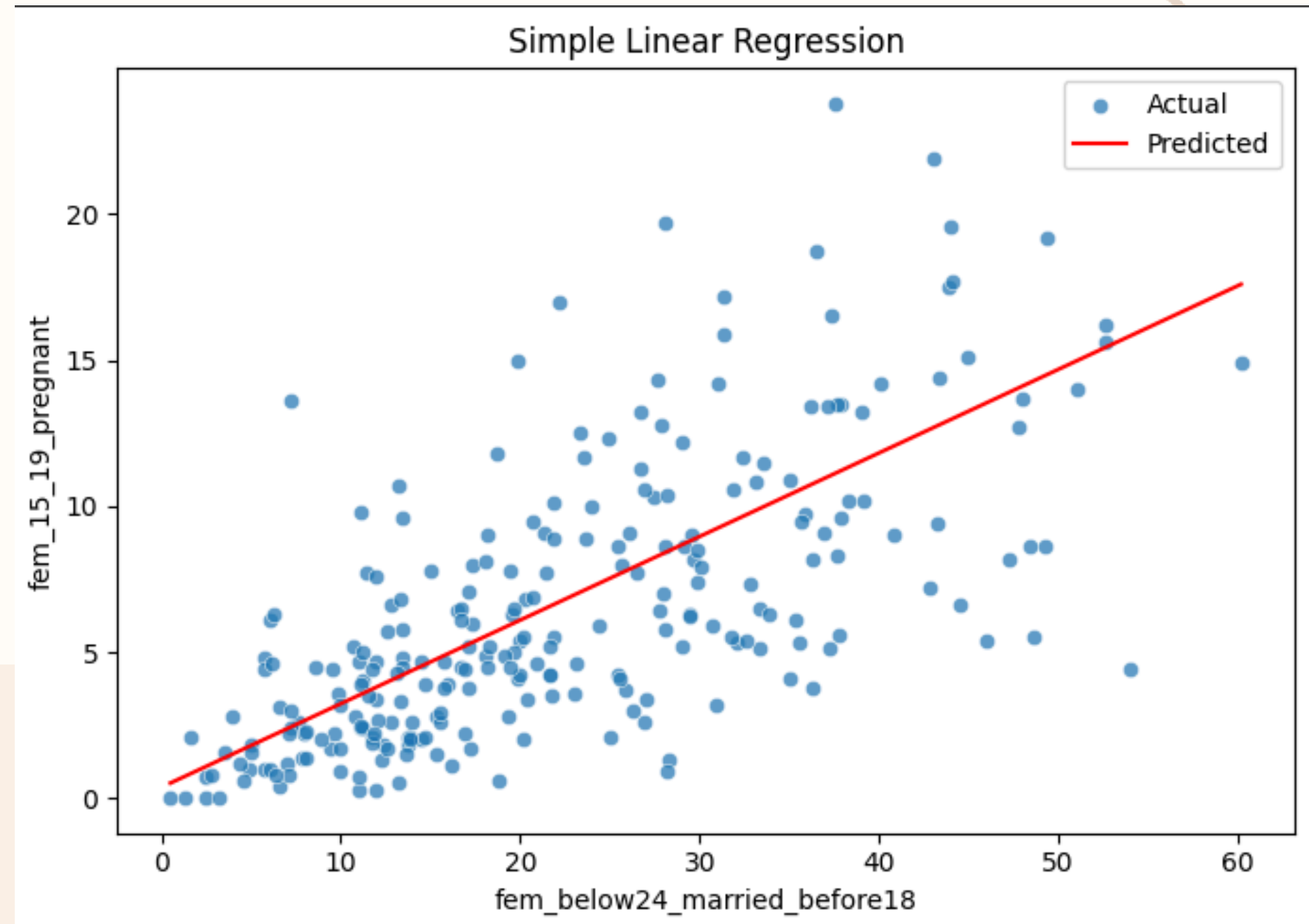
### Summary of Results:

Intercept: 0.37 | Slope: 0.29

For every 1% rise in early marriage, teen pregnancy increases by  $\approx 0.29\%$ .

$R^2 = 0.447$ , meaning early marriage explains  $\sim 45\%$  of variation in teen pregnancy.

RMSE = 3.5, showing moderate prediction accuracy.





# Analytical Models and Methodology Overview

## MULTIPLE LINEAR REGRESSION

A multiple linear regression was applied to predict teenage pregnancy (%fem\_15\_19\_pregnant) using five socio-health predictors:

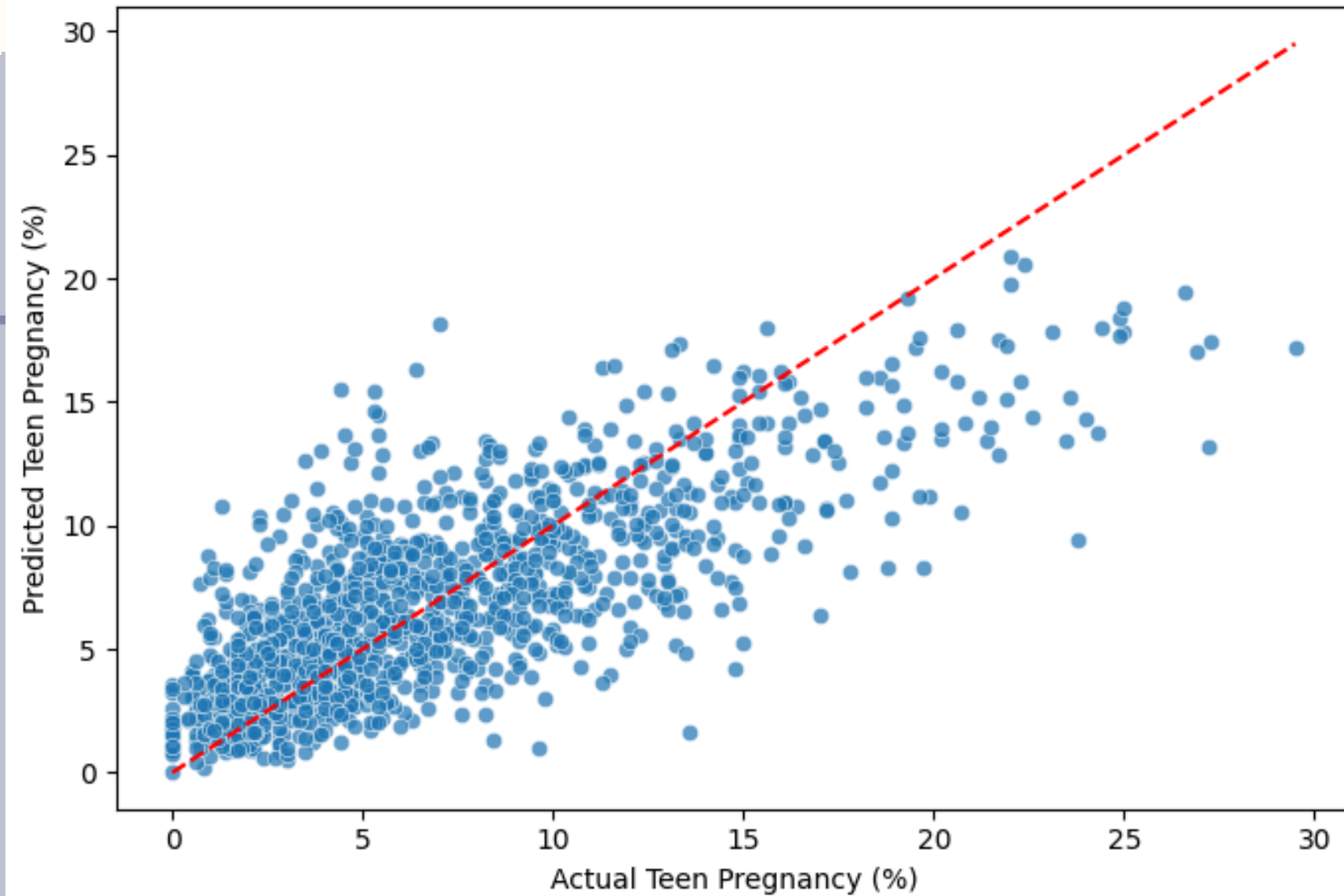
- early marriage rate (fem\_below24\_married\_before18)
- contraceptive use (fp\_pill)
- young population percentage (pop\_below\_15)
- child underweight rate (child\_5\_underweight)
- anaemia in pregnant women (wom\_preg\_15\_49\_anaemic)

## Summary of Results:

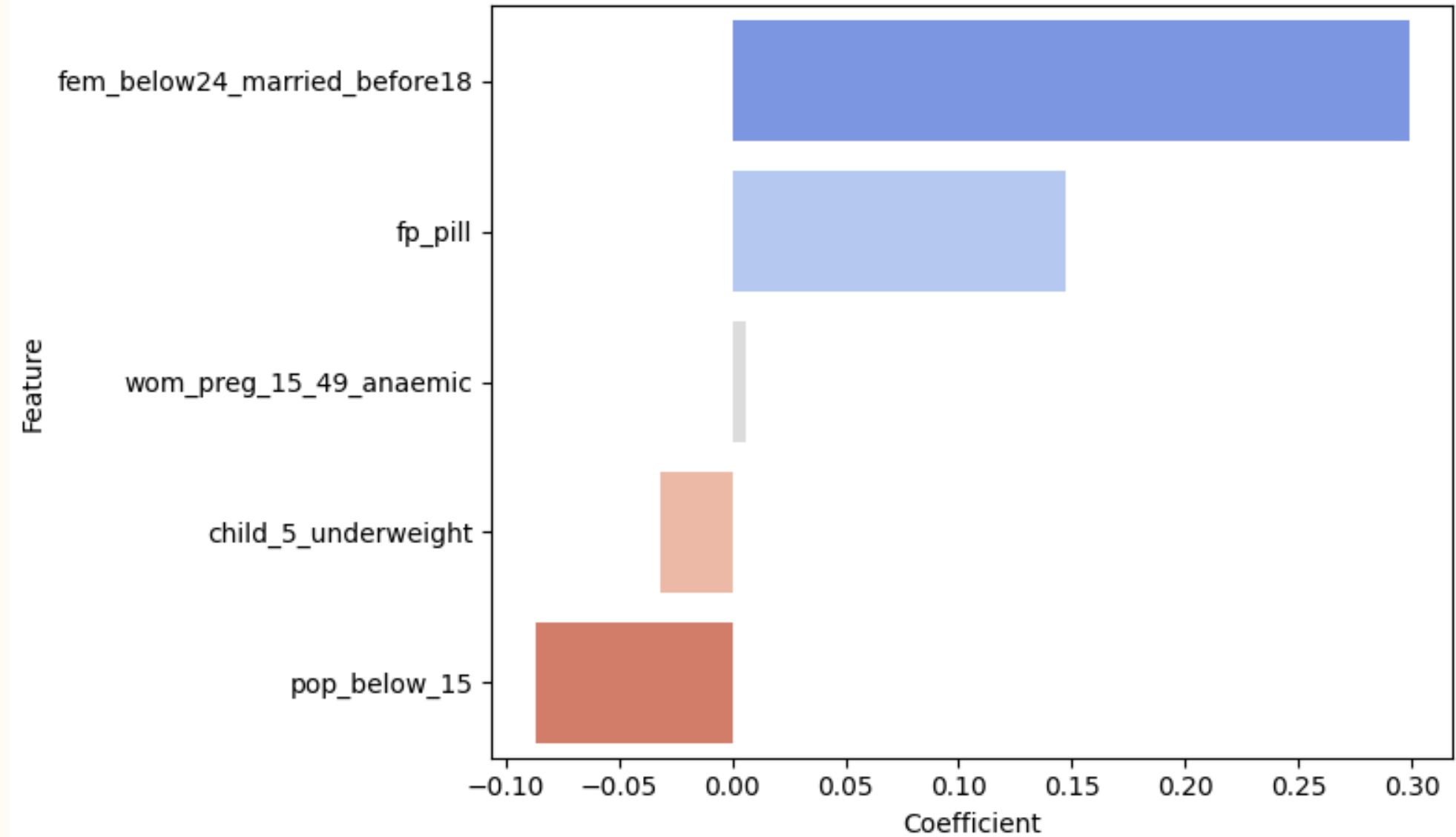
- Model performance improved notably compared to the simple regression:
- $R^2$  increased from 0.447  $\rightarrow$  0.587, showing a better overall fit (~30% improvement).
- RMSE decreased from 3.50  $\rightarrow$  3.30, meaning predictions became more accurate.
- The early marriage variable remained the strongest positive predictor.

# Graphs for Multiple Linear Regression

Multiple Linear Regression: Actual vs Predicted



Feature Importance - Multiple Linear Regression Coefficients





# Behavioral & Health Risk Clustering (K-Means)

K-Means clustering on behavioral and health indicators formed 6 distinct district groups, revealing strong links between tobacco/alcohol use and disease burden across India.

## **Purpose:**

To group districts based on men's and women's behavioral health indicators.

## **Variables Used:**

- Men's health behaviors: tobacco use (tobaco\_men\_15), alcohol use (alcohol\_men\_15), blood pressure severity (men\_bp\_sev)
- Women's behaviors: tobacco use (tobaco\_women\_15), alcohol use (alcohol\_women\_15)
- Disease indicators: oral cancer (oral\_cancer), cervical cancer (cerv\_cancer)

Cluster 0 – Very Low Risk

Cluster 1 – High Male Tobacco–Alcohol Use

Cluster 2 – Extreme Substance Use Both

Genders

Cluster 3 – Moderate Behavior, High Female Cancer

Cluster 4 – Severe Cancer Burden

Cluster 5 – Moderate Behavioral Risk

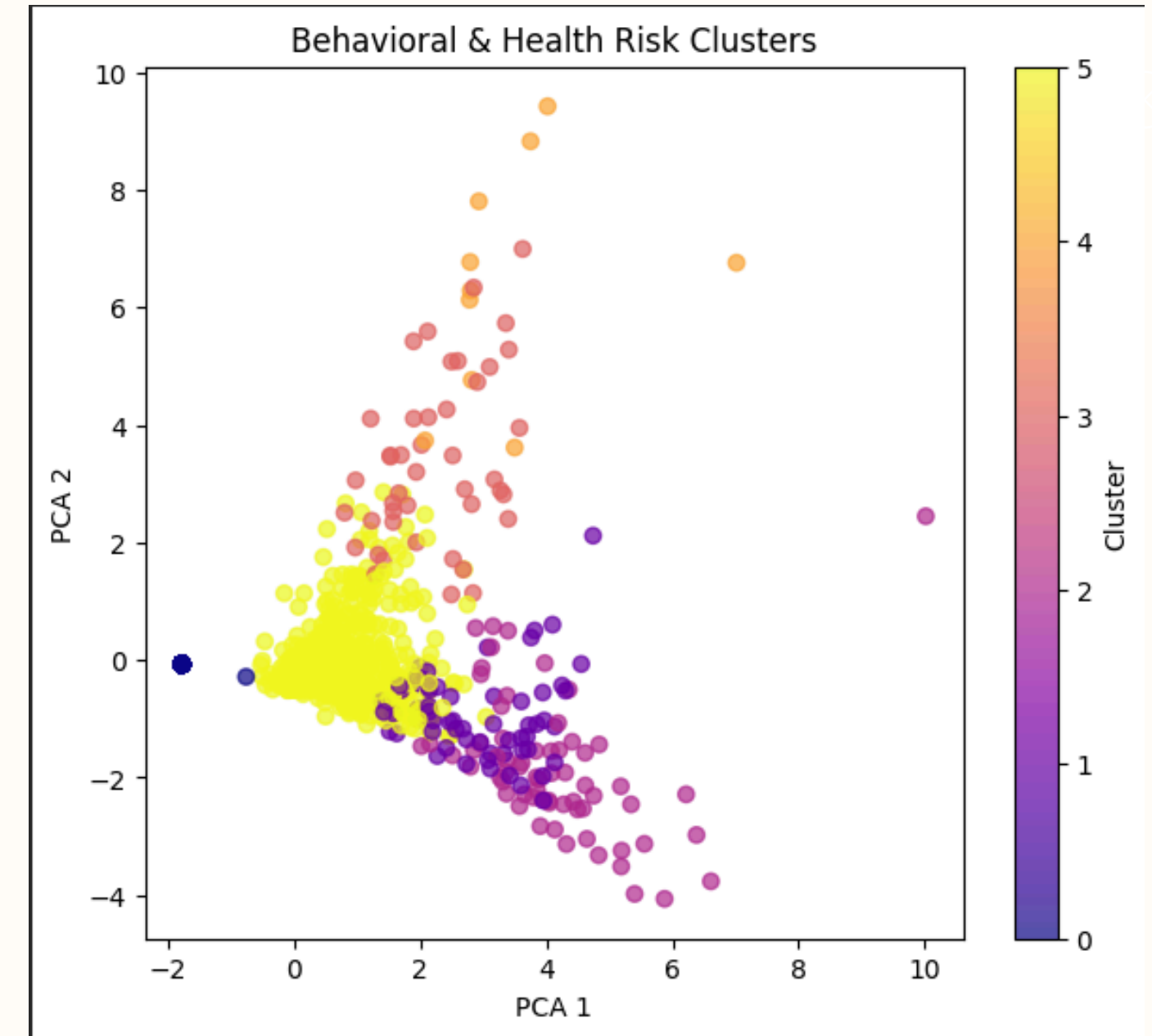
# Graphs

```
cluster_centers_b = pd.DataFrame(  
    scaler.inverse_transform(kmeans_b.cluster_centers_),  
    columns=features_behavioral  
)  
print(cluster_centers_b.round(2))
```

|   | tobacco_men_15 | alcohol_men_15 | men_bp_sev | tobacco_women_15 |
|---|----------------|----------------|------------|------------------|
| 0 | 0.06           | 0.00           | 0.00       | 0.00             |
| 1 | 59.23          | 27.91          | 5.84       | 40.36            |
| 2 | 50.16          | 49.07          | 7.75       | 20.75            |
| 3 | 21.27          | 30.79          | 7.67       | 5.18             |
| 4 | 30.42          | 26.42          | 7.23       | 10.09            |
| 5 | 39.24          | 18.63          | 5.71       | 7.83             |

|   | alcohol_women_15 | oral_cancer | cerv_cancer |
|---|------------------|-------------|-------------|
| 0 | 0.00             | 0.00        | 0.00        |
| 1 | 2.76             | 0.44        | 1.50        |
| 2 | 19.60            | 0.53        | 0.84        |
| 3 | 1.70             | 1.80        | 9.77        |
| 4 | 0.97             | 10.20       | 5.29        |
| 5 | 0.95             | 0.48        | 0.92        |



These very low p-values confirm the clusters are statistically distinct — not random groups. It means behavioral patterns and disease outcomes differ significantly across clusters.

```
oral_cancer: F = 545.52, p < 0.001  
cerv_cancer: F = 501.47, p < 0.001  
men_bp_sev:  F = 707.37, p < 0.001
```

# Women's Empowerment Clustering (K-Means)

## Purpose:

To identify regional patterns in education, early marriage, contraception, anaemia, and teen pregnancy.

## Variables:

female schooling, early marriage, teen pregnancy, contraceptive use, anaemia.

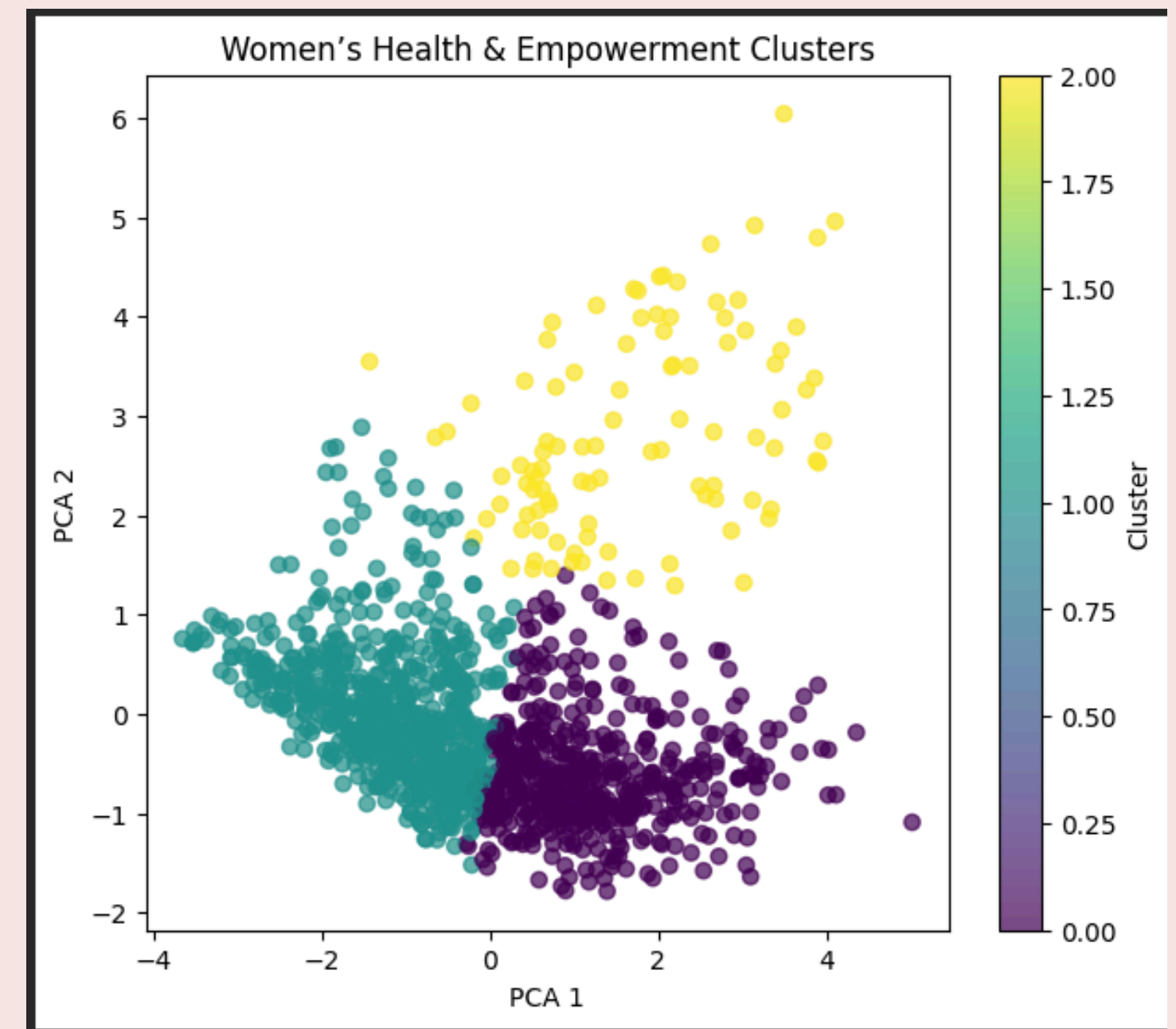
Clusters Found: 3

## Findings:

- Cluster 0-Transitional: Mid-level education, high anaemia, low contraception.
- Cluster 1 -Empowered & Healthy: High education, low early marriage, low anaemia.
- Cluster 2 – Educated but At-Risk: High education but high early marriage & teen pregnancy.

## Conclusion:

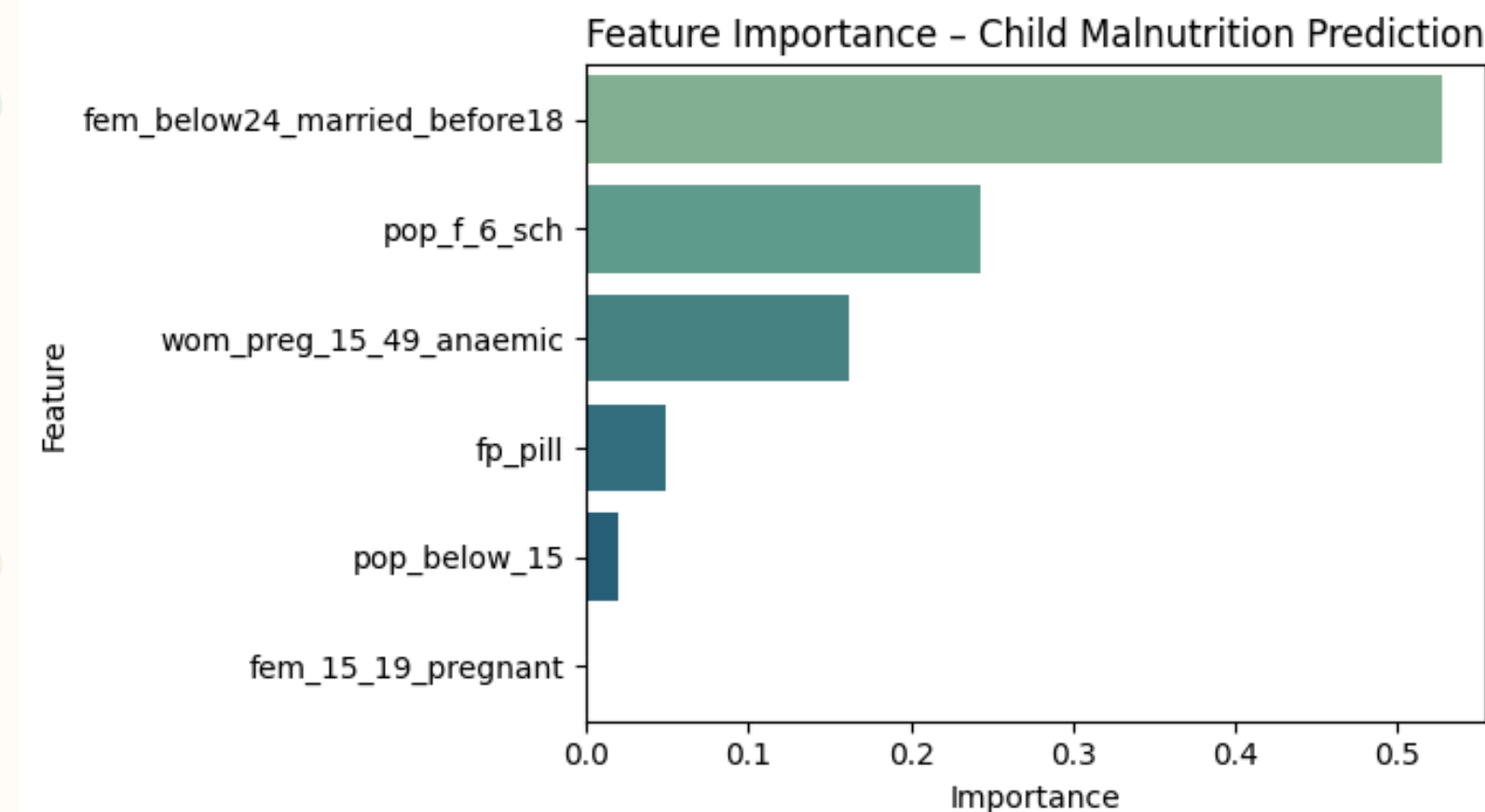
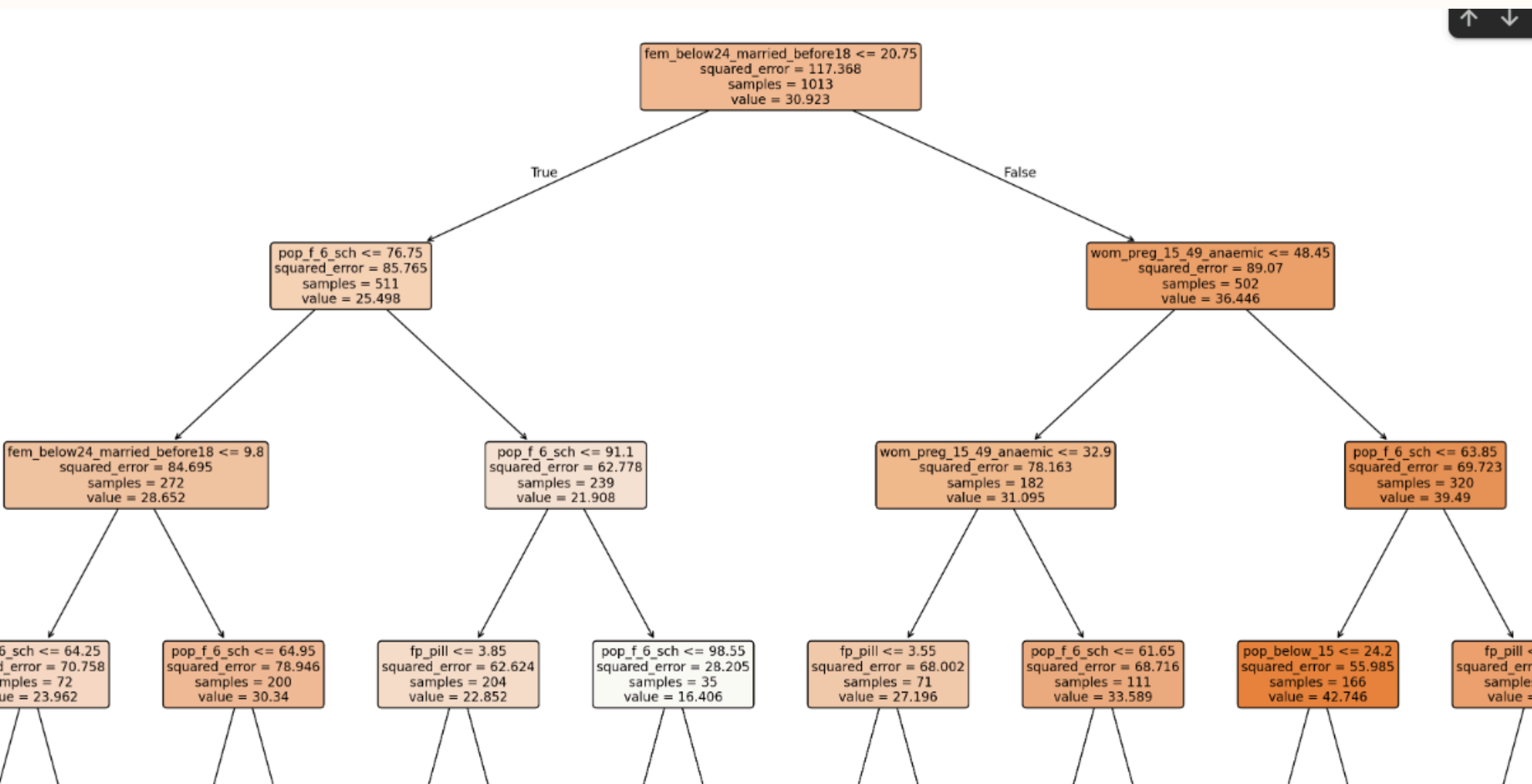
Education alone is not enough true empowerment requires health access and social change together.



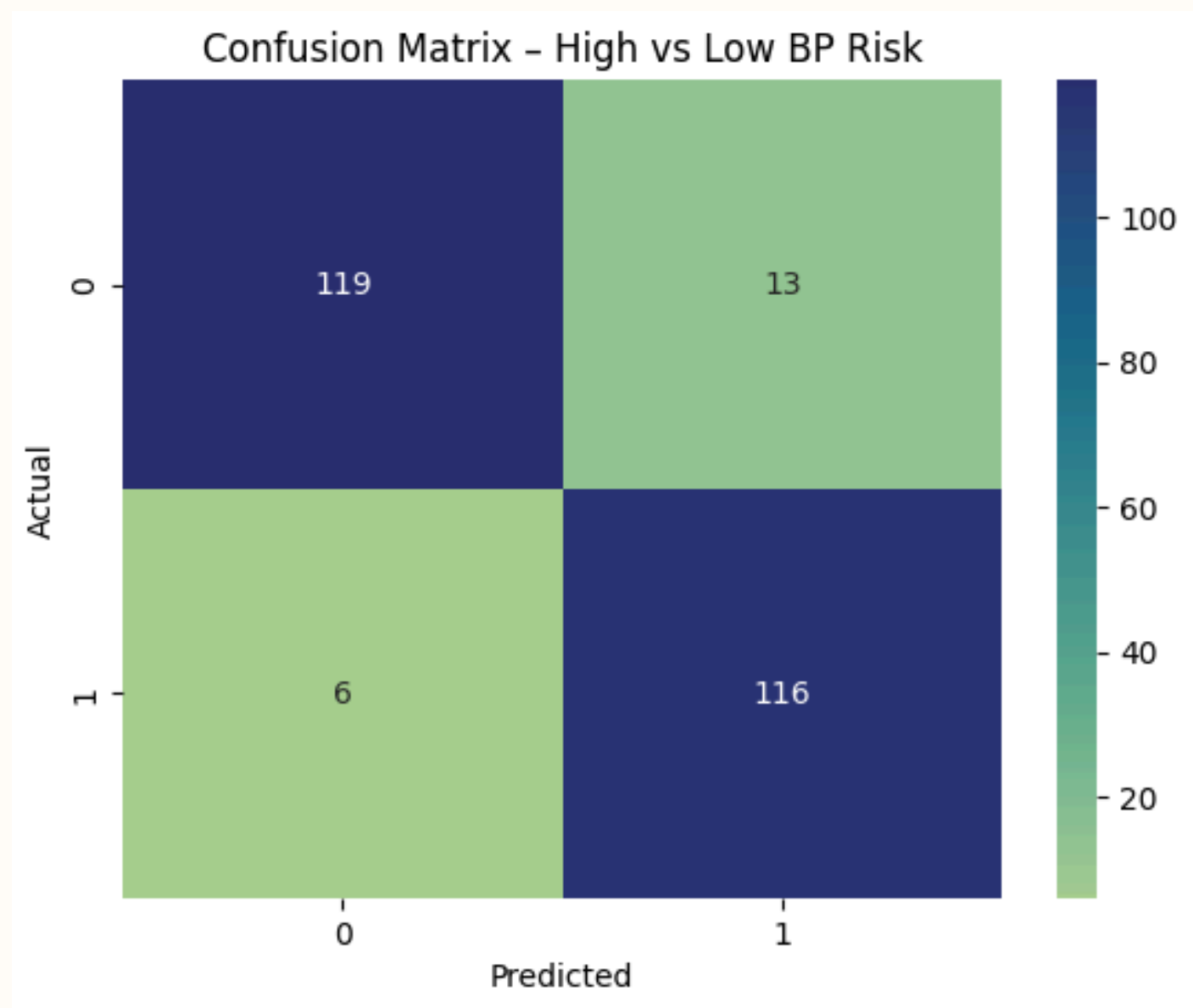
# DECISION TREE

To predict the percentage of underweight children under age 5 using a combination of women's health, education, and demographic indicators.

$$R^2 \approx 0.65, \text{RMSE} \approx 3.4$$



# LOGISTIC REGRESSION



To classify districts into High-Risk or Low-Risk zones for men's severe blood pressure (BP) using their lifestyle and health factors.

| Logistic Regression - Men's High BP Risk |           |        |          |         |
|------------------------------------------|-----------|--------|----------|---------|
| Accuracy: 0.925                          |           |        |          |         |
|                                          | precision | recall | f1-score | support |
| 0                                        | 0.95      | 0.90   | 0.93     | 132     |
| 1                                        | 0.90      | 0.95   | 0.92     | 122     |
| accuracy                                 |           |        | 0.93     | 254     |
| macro avg                                | 0.93      | 0.93   | 0.93     | 254     |
| weighted avg                             | 0.93      | 0.93   | 0.93     | 254     |

Model achieved 92.5% accuracy — male tobacco and alcohol use strongly predict hypertension risk among men.



# Ground Realities Reflected

6 out of 10

Women aged below 24 were married before 18

1 out of 3

Women (aged 15–49) suffer from anaemia

1 out of 5

Men consume alcohol regularly

3 out of 10

Teenage girls (15–19 years) are pregnant

All statistics are derived from the National Family Health Survey (NFHS-5) data used for model analysis.





Thank you!